

```
# ## Import Libraries

# The necessary libraries for data manipulation, visualization,
# feature engineering, model training and evaluation are imported.

import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import LabelEncoder

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import (
    confusion_matrix,
    ConfusionMatrixDisplay,
    accuracy_score,
    precision_score,
    recall_score,
    classification_report
)
```

```
# The dataset contains passenger demographic information,
# flight details, service ratings and the target variable
# 'satisfaction'.

df = pd.read_csv("/content/airline_passenger_satisfaction")

df.head()

df.shape

df.info

df.describe()
```

	Age	Flight Distance	Seat comfort	Departure/Arrival time convenient	Food and drink	Gate location
count	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000
mean	39.427957	1981.409055	2.838597	2.990645	2.851994	2.990422
std	15.119360	1027.115606	1.392983	1.527224	1.443729	1.305970
min	7.000000	50.000000	0.000000	0.000000	0.000000	0.000000
25%	27.000000	1359.000000	2.000000	2.000000	2.000000	2.000000
50%	40.000000	1925.000000	3.000000	3.000000	3.000000	3.000000
75%	51.000000	2544.000000	4.000000	4.000000	4.000000	4.000000
max	85.000000	6951.000000	5.000000	5.000000	5.000000	5.000000

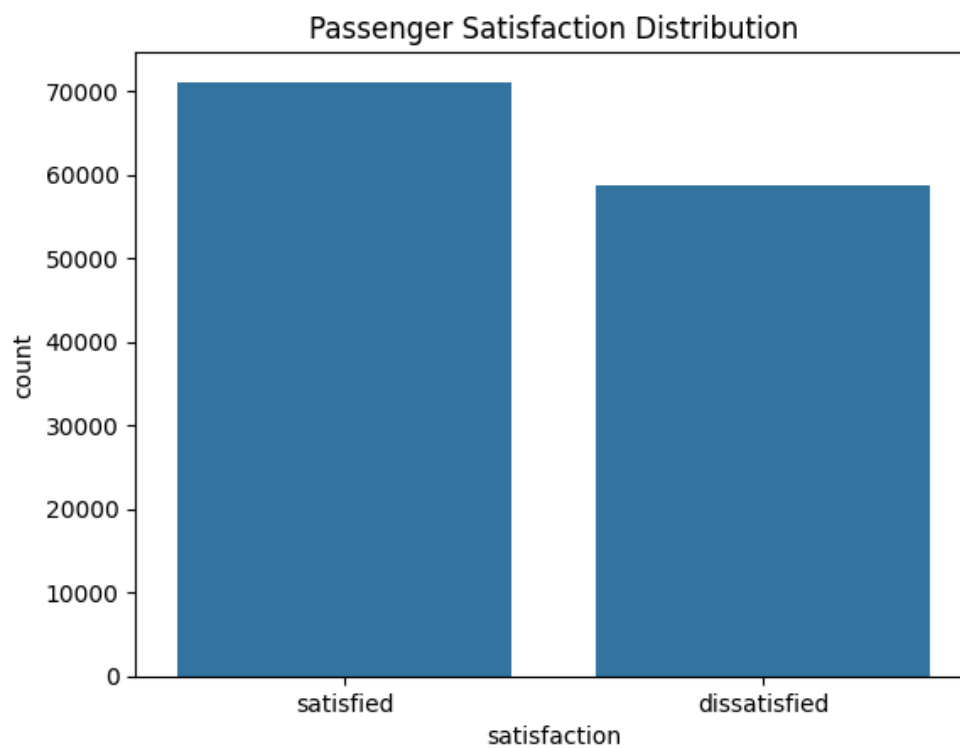
```
df['satisfaction'].value_counts()
```

	count
satisfaction	
satisfied	71087
dissatisfied	58793

dtype: int64

```
sns.countplot(data=df, x='satisfaction')
```

```
plt.title("Passenger Satisfaction Distribution")  
plt.show()
```



```
df.isnull().sum()
```

	0
satisfaction	0
Customer Type	0
Age	0
Type of Travel	0
Class	0
Flight Distance	0
Seat comfort	0
Departure/Arrival time convenient	0
Food and drink	0
Gate location	0
Inflight wifi service	0
Inflight entertainment	0
Online support	0
Ease of Online booking	0
On-board service	0
Leg room service	0
Baggage handling	0
Checkin service	0
Cleanliness	0
Online boarding	0
Departure Delay in Minutes	0
Arrival Delay in Minutes	393

dtype: int64

```
df.fillna(method='ffill', inplace=True)
```

```
/tmp/ipykernel_3677/4002983762.py:1: FutureWarning: DataFrame.fillna with 'method' is deprecated a
df.fillna(method='ffill', inplace=True)
```

```
df.dropna(inplace=True)
```

```
df.drop(columns=['ID', 'Unnamed: 0'], inplace=True, errors='ignore')
```

```
df.select_dtypes(include='object').columns
```

```
Index(['satisfaction', 'Customer Type', 'Type of Travel', 'Class'], dtype='object')
```

```
encoder = LabelEncoder()
```

```
for column in df.select_dtypes(include='object').columns:
    df[column] = encoder.fit_transform(df[column])
```

```
df.head()
```

	satisfaction	Customer Type	Age	Type of Travel	Class	Flight Distance	Seat comfort	Departure/Arrival time convenient	Food and drink	Gate location
0	1	0	65	1	1	265	0	0	0	
1	1	0	47	1	0	2464	0	0	0	
2	1	0	15	1	1	2138	0	0	0	
3	1	0	60	1	1	623	0	0	0	
4	1	0	70	1	1	354	0	0	0	

5 rows × 22 columns

#The predictor variables are separated from the target variable.

```
X = df.drop('satisfaction', axis=1)
```

```
y = df['satisfaction']
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
model = LogisticRegression(max_iter=1000)
```

```
model.fit(X_train, y_train)
```

/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html>

Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
n_iter_i = _check_optimize_result(
```

```
    LogisticRegression
    LogisticRegression(max_iter=1000)
```

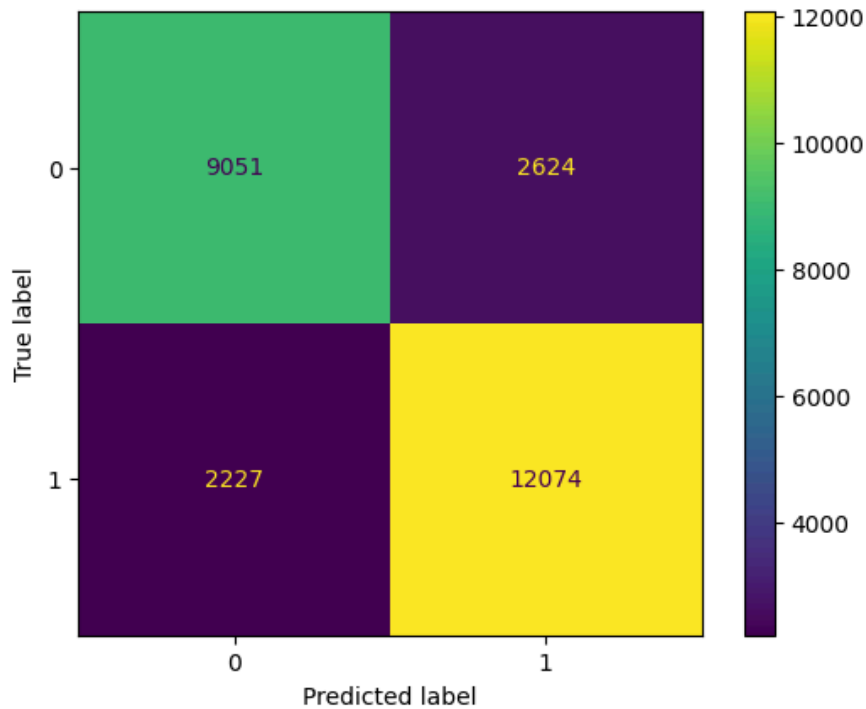
```
y_pred = model.predict(X_test)
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("Accuracy:", accuracy)
```

Accuracy: 0.813250692947336

```
cm = confusion_matrix(y_test, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot()
plt.show()
```



#Precision measures how many passengers predicted to be satisfied
#were actually satisfied.

```
precision = precision_score(y_test, y_pred)
print("Precision:", precision)
```

Precision: 0.8214723091577085

#Recall measures how many satisfied passengers
#the model successfully identified.

```
recall = recall_score(y_test, y_pred)
print("Recall:", recall)
```

Recall: 0.8442766240123069

```
print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	0.80	0.78	0.79	11675
1	0.82	0.84	0.83	14301

accuracy			0.81	25976
macro avg	0.81	0.81	0.81	25976
weighted avg	0.81	0.81	0.81	25976

```
coefficients = pd.DataFrame({
    'Feature': X.columns,
    'Coefficient': model.coef_[0]
})

coefficients.sort_values(by='Coefficient', ascending=False)
```

	Feature	Coefficient
10	Inflight entertainment	0.614825
12	Ease of Online booking	0.306585
13	On-board service	0.293590
5	Seat comfort	0.269646
16	Checkin service	0.202948
14	Leg room service	0.137987
18	Online boarding	0.121516
11	Online support	0.088638
8	Gate location	0.077966
19	Departure Delay in Minutes	0.000029
4	Flight Distance	-0.000374
20	Arrival Delay in Minutes	-0.004484
15	Baggage handling	-0.006654
1	Age	-0.026094
17	Cleanliness	-0.050812
9	Inflight wifi service	-0.178023
7	Food and drink	-0.184165
6	Departure/Arrival time convenient	-0.209371
2	Type of Travel	-0.696794
3	Class	-0.812874
0	Customer Type	-2.348408

Business Recommendations

Based on the Logistic Regression model, several actionable recommendations can help improve passenger satisfaction.

1. Improve the Online Boarding Experience

If **Online Boarding** is identified as one of the strongest positive predictors of satisfaction, the airline should invest in making the boarding process faster and more convenient. Enhancing the mobile boarding application, reducing waiting times, and providing real-time boarding notifications can improve the overall passenger experience.

2. Enhance In-flight Services

Features such as **Inflight Entertainment**, **Seat Comfort**, **Inflight Wi-Fi**, and **On-board Service** are expected to have a positive influence on customer satisfaction. The airline should continue investing in these services by improving entertainment options, ensuring reliable Wi-Fi connectivity, maintaining comfortable seating, and providing high-quality customer service during flights.

3. Minimize Flight Delays

If **Departure Delay** and **Arrival Delay** have negative coefficients in the model, reducing operational delays should be a strategic priority. When delays are unavoidable, proactive communication, timely updates, and appropriate customer compensation can help reduce dissatisfaction.

4. Target Service Improvements to Passenger Segments

If variables such as **Class**, **Type of Travel**, or **Customer Type** significantly influence satisfaction, the airline should develop targeted strategies for passenger groups with lower predicted satisfaction. Personalized offers, loyalty rewards, and tailored service improvements can increase customer retention and overall satisfaction.

8. Limitations and Future Work

This study has several limitations. First, Logistic Regression assumes a linear relationship between the predictor variables and the log-odds of passenger satisfaction, which may not capture more complex relationships within the data. Second, several variables in the dataset are based on passenger survey responses and may contain subjective opinions or response bias. Additionally, the model was evaluated using a single train-test split, and performance may vary with different data partitions.

For future work, more advanced machine learning algorithms such as Random Forest, Gradient Boosting, or XGBoost could be explored and compared with Logistic Regression to determine whether they provide better predictive performance. Furthermore, the final model could be integrated into the airline's customer relationship management system to identify passengers at risk of dissatisfaction and support data-driven service improvement initiatives.